

Title:

VitalWave: An End-to-End Open-Source High-Frequency Wearable Platform for Digital Health

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Abstract (in paragraph form):

Digital biomarkers derived from wearable devices have strong potential to support continuous, real-time health monitoring, but many commercial systems provide only low-frequency, preprocessed outputs and limited access to raw signals, restricting their utility for research and machine learning. To address this gap, we developed VitalWave, an end-to-end open-source wearable platform for high-frequency digital health data collection and analysis. VitalWave integrates a multimodal wearable device with an iOS mobile application, secure cloud storage, and a web-based dashboard for visualization and downstream analysis. The device captures raw physiological and motion data, including photoplethysmography, accelerometry, gyroscope, magnetometer, humidity, and temperature signals, at up to 100 Hz, while supporting real-time Bluetooth streaming and offline logging. Initial system evaluation using 10-minute walking trials assessed signal quality, data transmission, cloud synchronization, and dashboard functionality. VitalWave successfully streamed and stored high-quality multimodal data with no reported packet loss, produced physiologically relevant heart rate estimates compared with a Polar H10 reference, and supported automated cloud processing and web-based visualization. Battery life exceeded three days under standard usage. These findings demonstrate the feasibility of VitalWave as a low-cost, research-grade platform for digital biomarker development and wearable health studies. Future work will focus on broader validation, additional sensor integration, and expanded machine learning applications.

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